WORKED SOLUTION

# STEP III 2007, Question 2

Pure • Binomial Power Series

**STEP III 2007, Question 2**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q2

## The key idea

Everything flows from one binomial series. Once we know the coefficients of $(1-4x)^{-1/2}$, differentiating the series pulls down a factor $n$ (turning $(n!)^2$ into $n!\,(n-1)!$), and integrating it divides by $n+1$ (turning $(n!)^2$ into $n!\,(n+1)!$). So each later sum is the derivative or the integral of the first series, evaluated at a well-chosen point inside $|x| < \frac{1}{4}$.

## Solution

### Part (i)

Multiply and divide the product of the odd numbers by the missing even numbers $2\cdot 4\cdot 6\cdots(2n)$:

$$1\cdot 3\cdot 5\cdots(2n-1) = \frac{1\cdot 2\cdot 3\cdot 4\cdots(2n-1)(2n)}{2\cdot 4\cdots(2n)} = \frac{(2n)!}{2^n\,(1\cdot 2\cdots n)} = \frac{(2n)!}{2^n\,n!},$$

since each of the $n$ even numbers $2k$ contributes a factor 2 and a factor $k$.

For $|4x| < 1$, that is $|x| < \frac{1}{4}$, the binomial series gives

$$(1-4x)^{-1/2} = 1 + \sum_{n=1}^{\infty} \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2n-1}{2}\right)}{n!}\,(-4x)^n.$$

The numerator has $n$ factors, each negative, so it equals $(-1)^n\,\dfrac{1\cdot 3\cdots(2n-1)}{2^n}$. The $(-1)^n$ cancels the sign in $(-4)^n$, and $4^n/2^n = 2^n$, so the coefficient of $x^n$ is

$$\frac{2^n\cdot 1\cdot 3\cdots(2n-1)}{n!} = \frac{2^n}{n!}\cdot\frac{(2n)!}{2^n\,n!} = \frac{(2n)!}{(n!)^2},$$

using the first result. Hence, for $|x| < \frac{1}{4}$,

$$\frac{1}{\sqrt{1-4x}} = 1 + \sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2}\,x^n,$$

as required.

**Part (ii)**

A power series may be differentiated term by term inside its interval of convergence, here $|x| < \frac{1}{4}$. Differentiating both sides,

$$\frac{\mathrm{d}}{\mathrm{d}x}(1-4x)^{-1/2} = -\tfrac{1}{2}(1-4x)^{-3/2}\cdot(-4) = 2(1-4x)^{-3/2},$$

while on the right the constant disappears and $x^n$ becomes $nx^{n-1}$. Since $n/(n!)^2 = 1/(n!\,(n-1)!)$,

$$2(1-4x)^{-3/2} = \sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n-1)!}\,x^{n-1}.$$

Multiplying by $x$ gives the powers $x^n$ that we want:

$$\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n-1)!}\,x^n = \frac{2x}{(1-4x)^{3/2}} \qquad (|x| < \tfrac{1}{4}).$$

Now put $x = \frac{6}{25}$, which is allowed because $\frac{6}{25} < \frac{6}{24} = \frac{1}{4}$. Then $1-4x = \frac{1}{25}$, so $(1-4x)^{3/2} = \frac{1}{125}$, and

$$\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n-1)!}\left(\frac{6}{25}\right)^n = 2\cdot\frac{6}{25}\cdot 125 = 60.$$

**Part (iii)**

**Ask yourself** The factorials are now $(n+1)!\,n!$ rather than $(n!)^2$. What operation on the series produces that?

We need to divide the $n$th coefficient by $n+1$, since $(n+1)! = (n+1)\,n!$. Integrating $x^n$ does exactly this: $\int_0^x t^n\,\mathrm{d}t = \dfrac{x^{n+1}}{n+1}$. Term-by-term integration of a power series is also valid inside its interval of convergence, so for $|x| < \frac{1}{4}$ we integrate the series of part (i) from 0 to $x$:

$$\int_0^x (1-4t)^{-1/2}\,\mathrm{d}t = \Big[-\tfrac{1}{2}(1-4t)^{1/2}\Big]_0^x = \frac{1-\sqrt{1-4x}}{2},$$

and so

$$\frac{1-\sqrt{1-4x}}{2} = x + \sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n+1)!}\,x^{n+1}.$$

Next we match the powers in the target sum. Since $3^{2n} = 9^n = 9^{n+1}/9$,

$$\frac{2^{n+1}}{3^{2n}} = 9\left(\frac{2}{9}\right)^{n+1},$$

so the target sum is 9 times our series at $x = \frac{2}{9}$. This is a legitimate value, since $\frac{2}{9} < \frac{2}{8} = \frac{1}{4}$. There, $1 - 4x = \frac{1}{9}$ and $\sqrt{1-4x} = \frac{1}{3}$, giving

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{2^{n+1}(2n)!}{3^{2n}(n+1)!\,n!} &= 9\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n+1)!}\left(\frac{2}{9}\right)^{n+1} \\ &= 9\left(\frac{1-\frac{1}{3}}{2} - \frac{2}{9}\right) = 9\left(\frac{1}{3} - \frac{2}{9}\right) = 1,\end{aligned}$$

as required. Notice that the leading term $x$ (the integral of the constant 1) has to be subtracted, because the sum starts at $n = 1$.

## Checks

- In (i), $n = 2$ gives $1 \cdot 3 = 3$ and $\frac{4!}{2^2\,2!} = 3$; the series begins $1 + 2x + 6x^2 + 20x^3 + \cdots$, which matches squaring: $(1 + 2x + 6x^2)^2(1 - 4x) = 1 + O(x^3)$.
- Summing the first 3000 terms of (ii) with exact fractions gives $60.000\ldots$, and the first 400 terms of (iii) give $1.000\ldots$ to machine precision.
- In (iii) the first term ($n = 1$) is $\frac{4\cdot 2}{9\cdot 2} = \frac{4}{9}$, already a good share of the total 1; the terms then decrease roughly like $(8/9)^n$.

## Common slips

- Losing the signs in the binomial coefficient: there are $n$ negative factors in the numerator and $n$ in $(-4)^n$, so they cancel exactly.
- Forgetting to check that $\frac{6}{25}$ and $\frac{2}{9}$ lie inside $|x| < \frac{1}{4}$; outside it the series diverge.
- In (iii), forgetting the $x$ term from integrating the constant 1, which would give the answer 3 instead of 1.

## Beyond STEP

The numbers $\frac{(2n)!}{n!\,(n+1)!} = \frac{1}{n+1}\binom{2n}{n}$ that appear in part (iii) are the Catalan numbers $1, 1, 2, 5, 14, \ldots$, which count bracketings, binary trees and many other structures, and $\frac{1-\sqrt{1-4x}}{2x}$ is their generating function. The facts we used without proof, that a power series can be differentiated and integrated term by term strictly inside its radius of convergence, are standard theorems of first-year analysis; see Abbott, *Understanding Analysis*.

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